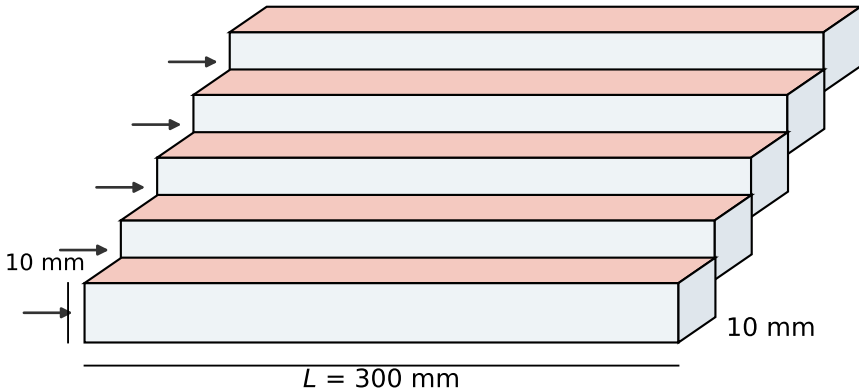


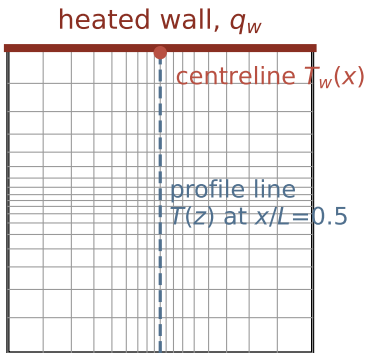
(a) cold plate heat exchanger domain uniform $q_w = \dot{Q}/A_{\text{heated}} = 627 \text{ W m}^{-2}$



inlet: T_{in} , $\dot{m}$ (He: mass-flow BC; LN₂: velocity BC); outlet: fixed p

other walls adiabatic and no slip; five square ducts, $D_h = 10 \text{ mm}$, pitch 18 mm
heated area 0.015 m^2 (top walls), flow area $5 \times 10^{-4} \text{ m}^2$

(b) duct cross-section: mesh and sampling lines



18×18 cells per duct
two-sided grading, ratio 6
first cell 0.19 mm
180 cells along L
 2.9×10^5 cells in total